

5.3.1

$A_f = \frac{A}{1+A_0F}$ 是一个单调函数 (当 F 一定)

$$A_{f1} = \frac{100}{1+100 \times 0.05} = 16.67 \quad A_{f2} = \frac{200}{1+200 \times 0.05} = 18.18$$

$\therefore$ 范围是 16.67 - 18.18.

5.3.2

$$A_f = \frac{A_0}{1+A_0F} \quad \frac{dA_f}{A_f} = \frac{1}{1+A_0F} \frac{dA_0}{A_0}$$

$$\therefore 1\% = \frac{1}{1+A_0F} \cdot 20\% \quad \therefore 1+A_0F = 20$$

$$\therefore A_f = \frac{A_0}{20} \quad A_0 = 20A_f = 2000 \quad F = \frac{20-1}{A_0} = 0.0095$$

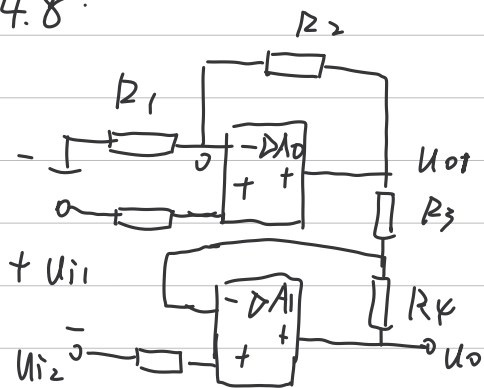
5.4.5

$$U_- = U_+ \quad U_+ \approx 0V \quad \therefore U_- = 0V$$

$$I = I_1 + I_2 + I_3 = \frac{U_1}{R_1} - \frac{U_2}{R_2} + \frac{U_3}{R_3} = \frac{3}{50} \times 10^{-3} A$$

$$U_0 = -IR_f = -\frac{3}{50} \times 10^{-3} \times 100 \times 10^3 V = -6V$$

5.4.8



$$\frac{U_-}{R_1} = \frac{U_{o1} - U_-}{R_2} \quad \therefore U_{o1} = 2U_-$$

$$U_- = U_{i1}$$

对第二个运放

$$\frac{U_{o1} - U_{i2}}{R_3} = \frac{U_{i2} - U_0}{R_4} \quad \therefore U_{o1} + U_0 = 2U_{i2}$$

$$\therefore U_0 = 2(U_{i2} - U_{i1}) = -2U_{i1}$$

